

Lab 9

This is the final lab, and is an exploration of data that was collected as part of a dissertation project. In this project, the researcher was interested in what variables predicted religious congregants trust of pastors in Southern Baptist congregations. The Southern Baptist church structure is one where congregants pick their pastor with little oversight from a larger religious body, in contrast to many other denominations where pastors are assigned by a larger religious structure. This type of loosely coupled organization is one where a pastor has less authority as a leader and must lead more by consensus of the congregants than authority from a structure.

In this study, the researchers measured several variables of interest. The most important of those are the ones at the end of the qualtrics data scored by the researchers and are described below

- Q9 - gender of congregant, categorical variable with levels female or male
- Q10 - age of congregant: categorical variable with levels 18-24 years old, 25-34 years old, 35-44 years old, 45-54 years old, 55-64 years old, and 65+ years old
- Q11 - how long have you known your pastor: categorical variable with levels 0-1 year, 2-4 years, 5-8 years, and 8+ years
- Q12 through Q16, various questions about participants closeness with pastors
- SC0 - Competence - Perceptions of how competent a pastor is seen overall
- SC1 - Warmth - Perceptions of how warm a pastor is seen
- SC2 - Competence_preaching - Perceptions of how competent a pastor is seen in preaching
- SC3 - Consistency - Perceptions of how consistent a pastor is seen, answers of questions such as “My pastor follows through on what he says he will do.” or “My pastor leads the church and its ministries in a consistent direction.”
- SC4 - Trust - ratings of how much people trust their pastor

Step 1: You should load the tidyverse and qualtrics libraries as a chunk.

Step 2: Import your dataset as a data frame named `lab9`. You should do this as a chunk using the `readSurvey` command. Once you do this, you need to have 471 observations of 54 variables. If you have issues with this, see lab 2 and lab 7.5 about importing data from qualtrics.

Step 3: Run the chunk below to remove data generated from previewing the survey and people who did not complete the survey. After this, you should have 356 observations.

```
lab9 = filter(lab9, Status != "Survey Preview")
lab9 = filter(lab9, Finished == "TRUE")
```

Step 4: Do women or men trust their pastor more? Run the appropriate test and tell me whether there is a significant difference using an annotation.

Step 5: See if there is a correlation between competence in preaching and trust in a pastor. Is this significant?

Step 6: Is competence or warmth more important in trusting a pastor? Run an appropriate analysis for this (you should do two analyses), and let me know what you think.

Step 7: Use an ANOVA to see whether there is a difference in pastoral trust based on age of congregant (variable Q10). Use the `aggregate` command to display the mean trust for each age level.

Step 8: List another hypothesis that you can test based on this data, using a correlation, t-test, or ANOVA. List it out and test it using the appropriate test.